Status	Finished
Started	Sunday, 1 November 2026, 8:28 PM
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Duration	5 mins 37 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search
b. Binary search
c. Linear search ✓
d. Hashing

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search ends successfully ✓
c. The search continues in the right sublist
d. The search continues in the left sublist

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Quick search
c. Merge search
d. Bubble search

Question **4**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Interpolation search
b. Linear search ✓
c. Binary search
d. Hash search

Question **5**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 10 ✓
c. 100
d. 500

Question **6**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Searching ✓
c. Discovering
d. Mining

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Selection

c. Complexity

d. Insertion

Question **8**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)

b. O(1) ✓

c. O👎

d. O(n log n)

Question **9**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted

b. When the list is large and sorted

c. When the list is small and unsorted ✓

d. When the list is small and sorted

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.

b. Linear search divides the list into halves.

c. Linear search requires the list to be sorted.

d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Both (1) & (3)
- c. Hash search
- d. Binary search

Question **12**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n/2
- c. log n
- d. n-1

Question **13**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the right sublist ✓
- b. In the middle of the list
- c. In the left sublist
- d. At the beginning of the list

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is sorted ✓
- b. When the list is very small
- c. When the list contains duplicate elements
- d. When the list is unsorted

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. Binary search algorithm is not efficient when the data elements more than 1500 ✓

d. There must be a mechanism to access middle element directly